Pallet Cubing Process

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- Overview
- Process
  - Determining Pallet Type
  - Sorting of oLPNs using Break Criteria
  - Determining Pallet Cubing Method
  - Consolidate Pallet into an oLPN
  - Sequencing of oLPNs using Sequence Criteria
  - Additional Scenarios
- Rule Attributes Supported for Pallet Cubing
- Pallet Cubing Constraints
  - Look Ahead Parameter inside Pallet cubing Constraints

Overview

Pallet cubing (or double cubing) occurs after oLPN cubing (either via UOM cubing or cube-to-capacity) has been completed. Once oLPNs are created, WM uses similar functionality to then cube those oLPNs onto outbound pallets.

WM takes the oLPNs selected by the Pallet Cubing Criteria and sequences them based on the associated sequencing criteria. Similar to residual cubing, WM compares oLPN's estimated weight and volume to the capacity of the pallet type as specified in the criteria, to determine the oLPNs to be cubed onto the pallets. Also, like residual cubing, multiple algorithms are supported for different goal (sequencing, minimising number of pallets or minimising total volume). Configurable breaking rules are also supported.

In addition, WM ensures that the total weight, volume, and number of oLPNs exceed the minimums specified for the pallet type being used. If the minimums for the pallet are not exceeded, the oLPNs are removed from the pallet and are therefore put on no pallet.

Process

Pallet Cubing is performed if Pallet Cubing is defined as a pipeline step, and a Pallet Cubing Strategy is specified in the Order Planning Strategy.

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It is not mandatory to have the Pallet Cubing step after the Cubing step is performed and before the Work Release step is invoked. The Pallet Cubing can either be done before or after Routing and Consolidation depending on whether pallet Cubing determination are based on oLPN Ship Via or destination locations etc.
oLPN is the primary object processed by pallet cubing. This oLPN object fetched as part of service will have all the oLPN details based on which rules can be determined. This includes oLPN Header attributes and LPN Details. Once Pallet Cubing is invoked, the oLPNs belonging to the same Order Planning Run Id are considered as part of Pallet Cubing selection criteria using oLPN or oLPN Detail level rule conditions and the corresponding Pallet Type is associated to each of the oLPNs.
There are 3 main steps performed as part of Pallet Cubing logic.

Determining Pallet Type

Pallet Types will be defined in the oLPN Component. These Pallet Types are configured oLPN Type = Pallet. These pallet types will have configuration on the maximum and minimum threshold defined for weight, volume, and number of oLPNs which can be cubed. Please refer to Container Type configurations for more information.
The Pallet Cubing Criteria will determine the Pallet Type used for the list of oLPNs which satisfies a particular rule criterion. All the oLPNs corresponding to one Criterion will be sent for further processing.

Sorting of oLPNs using Break Criteria

This provides an option to ensure the oLPNs can be segregated based on certain break criteria configured. For e.g., if we have to ensure oLPNs only from the same Shipment or Ship Via are to be cubed onto a Pallet, the oLPNs are selected and sorted upfront before the next steps. Only oLPNs belonging to the same break criteria parameter values defined will be eligible for being palletized onto the same outbound Pallet.
The available break criteria parameters are defined in the document in 'Rule Attributes' section.

Determining Pallet Cubing Method

There are three methods support to double cube the oLPNs on to a pallet. These are:

1. Sequencing algorithm - This algorithm ensures that oLPNs are sequenced in terms of the created date time of oLPNs. The reason behind the logic is that the oLPN creation will match with the sequencing criteria defined on residual cubing. So, if residual cubing is defined with a location pick sequence, then the oLPNs will be created with the same sequence so that user can pick these oLPNs in sequence and pick onto a planned Pallet.
2. Minimising total number of Pallets - This algorithm is math based and ensures that all the oLPNs belonging to the same order planning run Id are cubed optimally to have least number of outbound Pallets created.
3. Minimising total volume of the Pallets - This algorithm is math based and ensures that the total volume of all the Pallets

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created as part of Pallet Cubing is minimal.

The above 3 cubing methods are very similar to the cubing methods available as part of residual cubing. In the Pallet Cubing logic, the dimensions of the oLPNs are passed as inputs and compared with the Pallet Size Type dimension fits and whether the weight and volume of the oLPNs cubed are less than the allowed weights defined within each Pallet Type.
Once the Pallets are created and the oLPNs determined, cubing are then persist these Pallets with details of oLPNs linked to the Pallet. The oLPNs are populated with Planned Pallet Id to indicate that these oLPNs should be picked or consolidated into the suggested pallets. Also, oLPN component should ensure that all the corresponding common header parameters of the oLPN should be passed on to the Pallet. For e.g., if all the oLPNs have the same Ship Via, then this information will be retained at the Pallet header information.

Consolidate Pallet into an oLPN

Pallet Cubing also has an additional feature to consolidate all the oLPNs on the Pallet into one oLPN. For e.g., if there were 10 oLPNs allocated to the Pallet, the oLPN records are consumed and the inventory within these oLPNs will be consolidated onto a bigger oLPN (which is physically the Pallet). This feature can be used if all the oLPNs are destined for the same Distribution Order. This is driven through a parameter in Pallet Cubing Criteria which indicates whether all oLPNs should be consolidated.
Consolidated oLPN creation can be controlled with the pallet cubing criteria parameter "Should the system assign a new pallet oLPN ID for the consolidated pallet LPN?". If the parameter is set to Yes, then the system creates a new oLPN ID by cloning the details from the consolidated bigger oLPN and olpn number is prefixed with the pallet prefix-001 to treat the new oLPN as a physical pallet in the warehouse. If the parameter is set to No, then the system consolidates all the oLPN details into a bigger oLPN among the list of oLPNs given to the pallet cubing.

Sequencing of oLPNs using Sequence Criteria

Sequencing of oLPNs on Pallet based on the user configurable attributes. Users can adjust the placing of oLPNs on the Pallet based on the operation and business requirements.

Additional Scenarios

Exception Scenarios	System behavior
Rule group defined for Pallet Cubing criteria but no matching criteria is found	No Pallet Cubing will be performed on the oLPN. **No error should be thrown** either and hence the wave will proceed.
If the remaining oLPNs falls below the minimum limit for Pallet size, and are hence de-palletized.	For each pallet created, WM compares its total weight, volume, and number of oLPNs with the minimum threshold values as specified in the pallet type. If any of the thresholds is not met, all its oLPNs are moved from the pallet. The oLPNs will remain standalone and un-palletized. The pallet in consideration should be the smallest size of the corresponding pallet type.
If rules are based out of oLPN Details, and not all oLPN Details fall under the rule defined.	Even if one oLPN Detail satisfies the rules defined, the oLPN is selected for Pallet Cubing Criteria.
oLPN is already Palletized as part of Pallet UOM Cubing logic.	These oLPNs which are physically Pallets being allocated, will not be considered for Pallet Cubing logic.

Rule Attributes Supported for Pallet Cubing

The Entities and corresponding attributes which are required to support pallet Cubing

1. Order
2. Order Line

3. OLPN
4. oLPNDetail(Only for break criteria)
5. Item
6. FirstLcoation
7. LastLocation
8. Facility
9. FacilityAddress

These entities will be used for rule evaluation to determine the Pallet Cubing Criteria.

Entity	Attributes for Rule Evaluation	Attributes for Sequencing Criteria	Attributes for Break Criteria
Order	orderId orderType merchandizingDepartment vendorId designatedModeId designatedCarrierId designatedServiceLevelId destinationAddress designatedShipVia designatedStaticRouteId customerId customerName hotOrder originFacilityId destinationFacilityId referenceShipmentId deliveryStartDateTime deliveryEndDateTime pickupStartDateTime pickupEndDateTime	orderId orderType merchandizingDepartment vendorId designatedModeId designatedCarrierId designatedServiceLevelId destinationAddress designatedShipVia customerId customerName originFacilityId destinationFacilityId designatedShipmentId	orderId orderType merchandizingDepartment vendorId designatedModeId designatedCarrierId designatedServiceLevelId destinationAddress designatedShipVia customerId customerName originFacilityId destinationFacilityId designatedShipmentId
Order Line	orderId orderLineId originalOrderId originalOrderLineId itemId productStatusId countryOfOrigin vasProcessType customerItemNbr totalMonetaryValue hazmat dangerousGoods priority assignedShipmentId assignedShipVia Allocated quantity Batch Number Description Ordered Quantity, Quantity UOM ID	originalOrderId originalOrderLineId description priority assignedShipmentId orderId orderLineId itemId productStatusId countryOfOriginId hazardousMaterial climateControlId assignedShipViaId orderedQuantity allocatedQuantity quantityUomId batchNumber	originalOrderId originalOrderLineId description priority assignedShipmentId orderId orderLineId itemId productStatusId countryOfOriginId hazardousMaterial climateControlId assignedShipViaId orderedQuantity allocatedQuantity quantityUomId batchNumber

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OLPN	olpnId originFacilityId destinationFacilityId orderId shipmentId shipViaId containerTypeId containerSizeId calculatedCutoffTimestamp destinationLocationId estimatedWeight estimatedVolume totalLpnQty length width height crossReferenceLpnId isSingleLine lpnCreationCode referenceLpnId externalShipmentId externalManifestId totalCharge chargeCurrency	olpnId originFacilityId destinationFacilityId orderId shipmentId shipViaId containerTypeId containerSizeId lpnCreationCode referenceLpnId externalShipmentId externalManifestId crossReferenceLpnId createdTimestamp estimatedWeight estimatedVolume height width length	olpnId originFacilityId destinationFacilityId orderId shipmentId shipViaId containerTypeId containerSizeId lpnCreationCode referenceLpnId externalShipmentId externalManifestId crossReferenceLpnId createdTimestamp estimatedWeight estimatedVolume height width length
Item	itemId storeDepartment catchWeightItem productClass protectionLevel commodityCode commodityClass Color Suffix Height Length Width Weight, Weight UOM Volume, Volume UOM, Description Critical Dimension1 Critical Dimension2 Critical Dimension3 Style Style suffix Season Season Year Freight Class	itemId description length width height volume weight season seasonYear style styleSuffix colorSuffix volumeUomId weightUomId climateControlId storeDepartment catchWeightItem productClass commodityCode nmfcFreightClassId criticalDimension1 criticalDimension2 criticalDimension3	itemId description length width height volume weight season seasonYear style styleSuffix colorSuffix volumeUomId weightUomId climateControlId storeDepartment catchWeightItem productClass commodityCode nmfcFreightClassId criticalDimension1 criticalDimension2 criticalDimension3

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firstLocation	locationId locationBarcode area aisle bay level zone position locationDepth locationWidth displayLocation locationHeight pickAllocationSequence pickAllocationZoneId locationLevel	locationId locationBarcode area aisle bay level zone position locationDepth locationWidth displayLocation locationHeight pickAllocationSequence pickAllocationZoneId locationLevel	locationId locationBarcode area aisle bay level zone position locationDepth locationWidth displayLocation locationHeight pickAllocationSequence pickAllocationZoneId locationLevel
lastLocation	locationId locationBarcode area aisle bay level zone position locationDepth locationWidth displayLocation locationHeight pickAllocationSequence pickAllocationZoneId locationLevel	locationId locationBarcode area aisle bay level zone position locationDepth locationWidth displayLocation locationHeight pickAllocationSequence pickAllocationZoneId locationLevel	locationId locationBarcode area aisle bay level zone position locationDepth locationWidth displayLocation locationHeight pickAllocationSequence pickAllocationZoneId locationLevel
facility	facilityId vendorId facilityName customerOwnedFacility	facilityId vendorId facilityName customerOwnedFacility	facilityId vendorId facilityName customerOwnedFacility
facilityAdress	address1 city state postalCode county country	address1 city state postalCode county country	address1 city state postalCode county country

Pallet Cubing Constraints

The concept of adding an additional pallet cubing constraint for each oLPN being considered for cubing - was introduced to bring more efficiency in the pallet cubing process for oLPNs, to ensure that certain oLPN categories or weight class does not exceed more than a specific percentage of weight or volume of the total Pallet size. This allows a more efficient way to cube multiple oLPN categories so that fewer pallets are used for oLPNs. It also supports putting limits on heavy good items to be cubed so that the users don’t encounter a safety concern and try to place more heavy items on a pallet beyond a specified threshold. This is applicable only if Pallet cubing is enabled with greedy algorithm in sequence.

For e.g. If there are multiple weight categories, like Heavy, Medium and Light goods of oLPNs, and there is a need to ensure that heavier oLPNs are always placed before the lighter goods, so, those can be sequenced accordingly. However, if we start cubing only the heavier oLPNs first, then it will not be an optimised pallet because the weight constraints associated with the pallet might be reached faster and would not utilise the remaining space available in the pallet to cube the lighter goods. Similarly, if we start

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cubing all the lighter oLPNs, then the volume constraints would be reached faster and hence the remaining available weight capacity is not utilised.

Example:

If we have 8 oLPNs with a mixed weight class of Heavy, Medium and Light that are sequenced in a descending weight and the Pallet weight capacity is 1000 lbs and volume capacity as 700 cuft.

Without Pallet cubing constraints- When the system goes through the regular Pallet cubing 3 pallets will be generated, 1st pallet with with oLPN1 and oLPN2, 2nd with oLPN3, oLPN4, oLPN5, oLPN6, oLPN7 and now since the weight and volume capacity is reached for both the pallets so a 3rd pallet is generated with one remaining oLPN8.

With Pallet cubing constraints- Now, if we define the Pallet constraint for Heavy as 80% of the total pallet weight and for Medium as 95% of the total pallet weight, then in that case, the system generates 2 Pallets, with the defined constraints i.e. 1st Pallet with oLPN1, oLPN2 and oLPN5, keeping in mind the weight and volume capacity which is reached for the 1st Pallet so another pallet will be generated for the remaining oLPN3, oLPN4, oLPN6, oLPN7 and oLPN8.

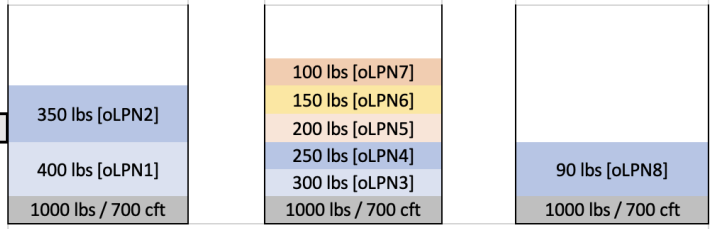
Sequenced by : oLPNEstimatedWeight DESC

Pallet Weight Capacity = 1000 lbs

Pallet Volume Capacity = 700 cuft

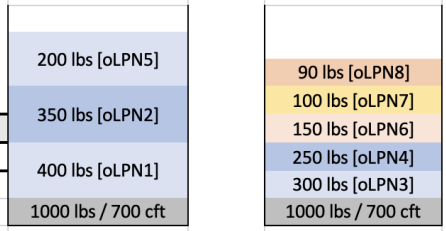
oLPN	WeightClass	Weight	Volume
OLPN1	Heavy	400 lbs	200 cuft
OLPN2	Heavy	350 lbs	250 cuft
OLPN3	Medium	300 lbs	250 cuft
OLPN4	Medium	250 lbs	150 cuft
OLPN5	Medium	200 lbs	100 cuft
OLPN6	Heavy	150 lbs	70 cuft
OLPN7	Medium	100 lbs	80 cuft
OLPN8	Light	90 lbs	70 cuft

Without Pallet Cubing constraints



With Pallet Cubing constraints

Pallet cubing LookAhead=10



Rule	Parameter	Value	LookAhead
WeightClass = Heavy	MaxCubingWeight	80%	0
WeightClass = Medium	MaxCubingWeight	95%	10

Look Ahead Parameter inside Pallet cubing Constraints

Within each Pallet cubing constraint for a specific rule, there is a look ahead parameter enabled. This is to ensure that if the capacity constraint is already reached for a certain category of oLPNs, the look ahead parameter restricts further looking ahead within the same category.

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